

Demonstrate the bouncing ball animation using macromedia Flash

Introduction

Macromedia Flash is a powerful tool for creating computer based animations. This lab demonstrate the process of animating a ball bouncing down a staircase using keyframes. This animation is created manually by adjusting the position of the frame by frame.

Objective

- To understand key-frame based animation in macromedia Flash
- To create a realistic bouncing ball effect using keyframe
- To manually adjust object positions for smooth animations
- To learn about frame rates and their impact on animation smoothness

Theory of computer Based animation

computer-Based animation is the process of creating moving images using digital tools and software. It involves generating sequence of images known as frames, that when played in succession create the illusion of motion.

Key concept includes

Keyframes: Define the start and end of motion. These are the frames where significant changes occur in an animation.

Frame rate: measure in frames per-second (FPS). It determines animation smoothness. A higher FPS result in a smoother animation.

Manual Frame-by-frame Animation: Unlike tweening, where movement is generated automatically, manual animation requires adjusting each frame individually.

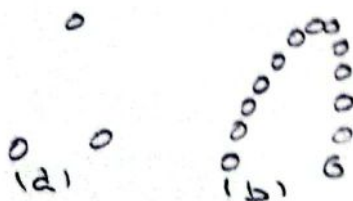


Fig: Linear interpolation of motion of ball



Fig Timing chart for ball bounce

Basic Steps for Computer-Based Animation

(a) Input process

At the initial stage of animation, graphical elements are drawn and placed on separate layers. Keyframes are inserted at points where the ball changes direction. Each keyframe represents a major position change.

(b) composition stage

The foreground (ball) and background (staircase) are combined. The ball's position is adjusted frame by frame to simulate bouncing. Careful alignment is required to ensure the motion appears natural.

c. In Between process

The animation of movement from one position to another requires intermediate frames between keyframes. The manual process ensures a smooth transition by adjusting the ball's positions slightly in each frame.

d) Adjusting motion and speed

By modifying the distance the ball moves between frames, different speeds of motion can be simulated. For eg. Larger gaps between frames can indicate faster motion, while smaller gaps create a slower effect.

Practical Implementation: Bouncing Ball animation

Steps to create Animation

(1) create a new flash document

- Set the stage size to 550 to 400 pixels
- Set frame rate to 12 fps for smooth movement

2 Draw the objects

- Use the shape tool to create a staircase
- Use the circle tool to create a ball

3 Insert Keyframes

- Right click on the timeline and insert keyframes at position where ball changes direction
- Each keyframe makes a movement when the ball bounces

(4) manually adjust object position

5 Test and Export

- Use the 'control > play' option to preview the animation
- Save and export animation as a SWF file

Analysis and Disussion

The bouncing ball animation demonstrates the importance of key framing in computer animation. During the animation process, several challenges were encountered such as:

- creating smooth transition between keyframes
- Adjusting timing to make the motion appear realistic
- controlling the frame rate to ensure smooth movement
- gradually reducing the bounce height to achieve a natural effect
- using easing techniques to make the motion look more natural
- maintaining a consistent path for the ball to ensure smooth animation.

Conclusion

This activity helped demonstrate how to create a bouncing ball animation using Macromedia Flash. Important animation concepts such as key framing, frame rate control, easing, and manual adjustments were applied to make the motion smooth and natural. Learning these basic concepts is useful for understanding animation fundamentals and for preparing for more advanced motion graphics.

